SOC Internship - Day 5

1. What are Logs?

Logs are records of events that happen inside a computer system, application, network, or device. They help administrators, developers, and Cybersecurity teams understand what happened, when it happened, and sometimes who caused it.

2. Why Logs are Important

Logs are important because they help us understand what is happening inside a computer system, network, or application. They act like evidence or records of activity

1. Troubleshooting Problems

Logs help identify errors and system failures.

Example:

- A website crashes
- Logs show the exact error message and time

This helps developers fix the issue faster.

2. Detecting Cyber Attacks

Security teams use logs to detect suspicious activities.

Examples:

- Multiple failed login attempts
- Malware activity
- Unauthorized access
- Unusual network traffic

A SOC analyst checks logs to identify threats early

3.Monitoring User Activity

Logs record what users do on a system.

Examples:

- Who logged in
- What files were accessed
- What changes were made

This helps with accountability and auditing

Incident Investigation

After a cyberattack or system failure, logs help investigators understand:

- What happened
- When it happened
- Who was involved

- How the attack occurred

Logs are like digital evidence in forensic investigations.

5. Improving System Performance

Logs help administrators monitor:

- CPU usage
- Memory problems
- Slow applications
- Server overload

This helps improve performance and stability.

3. Types of Logs

System Logs-> OS and hardware events

Authentication Logs-> Login activities

Application Logs -> Software/application events

Security Logs -> Security monitoring

Network Logs -> Network traffic

Web Server Logs -> Website access details

Database Logs -> Database activities

Firewall Logs -> Traffic filtering

Audit Logs -> User action tracking

Event Logs-> Windows system events

Class Activity: Analyze the Logs Below

Apr 10 10:17:45 Failed password for user1 from 192.168.1.10

Apr 10 10:18:02 Failed password for user1 from 192.168.1.10

Apr 10 10:18:15 Failed password for user1 from 192.168.1.10

Apr 10 10:18:30 Failed password for user1 from 192.168.1.10

Apr 10 11:05:10 Accepted password for admin from 203.0.113.45

Apr 10 11:06:22 Accepted password for admin from 203.0.113.45

Apr 10 12:12:11 Failed password for user2 from 192.168.1.20

Apr 10 12:12:15 Failed password for user2 from 192.168.1.20

Apr 10 12:15:30 Accepted password for user2 from 192.168.1.20

1. Which IP looks suspicious?
192.168.1.10

2. What type of attack is this?

Brute-Force Attack

3. Why do you think it is suspicious?

- Multiple failed login attempts happened within seconds
- All attempts targeted the same user account (`user1`)
- The attempts came from the same IP address
- Repeated password failures are a common sign of password guessing attacks